

The conjectured generating function of OEIS A063219, from the recurrence the entry states

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Abstract

OEIS A063219 states a linear recurrence as a fact and carries a rational generating function as an empirical conjecture. The conjecture follows from the recurrence by polynomial algebra alone: the recurrence fixes the denominator, the published terms fix the numerator exactly, and the generating function is then determined as a rational function rather than checked on finitely many coefficients. No model of the sequence and no reading of its name is required.

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1 What the entry says

The first terms are 4, 13, 21, 31, 39, 47, 57, 65, $\dots$, with offset 1. The entry states, as a fact,

$$a(n) = a(n-1) + a(n-3) - a(n-4) \text{ for } n \geq 5.$$

and separately records the conjecture

$$\text{Empirical g.f.: } -x^*(x^4 - 6x^3 - 8x^2 - 9x - 4) / ((x-1)^2(x^2 + x + 1)).$$

The stated line was taken from the entry's factual formulas only. A formula qualified by a finite range – "holds at least up to $n = 1000$ but is not known to hold in general" – is not a fact and is not used, nor is anything inside a conjectural block.

2 The argument

Write $b_k = a(1+k)$ and

$$D(x) = 1 - \sum_i c_i x^i = x^4 - x^3 - x + 1,$$

the polynomial of the stated recurrence, of degree 4. Let $B(x) = \sum_{k \geq 0} b_k x^k$.

The recurrence says $b_k = \sum_i c_i b_{k-i}$ for every $k \geq 5$. The coefficient of x^k in $D(x)B(x)$ is exactly $b_k - \sum_i c_i b_{k-i}$, so it vanishes for every $k \geq 5$ and

$$D(x)B(x) = P(x) = \sum_{k < 5} \left(b_k - \sum_i c_i b_{k-i} \right) x^k = -x^5 + 6x^4 + 8x^3 + 9x^2 + 4x,$$

a polynomial whose every coefficient is fixed by finitely many published terms. Therefore

$$B(x) = \frac{P(x)}{D(x)} = -\frac{x(x^4 - 6x^3 - 8x^2 - 9x - 4)}{(x-1)^2(x^2 + x + 1)},$$

which is the conjectured expression.

Theorem 1. *The conjectured generating function of OEIS A063219 is correct, given the recurrence the entry states.*

Proof. Immediate from the displayed identity: $DB = P$ holds coefficientwise, $D(0) = 1 \neq 0$ so D is invertible in the ring of formal power series, and $B = P/D$ follows. This is an identity of rational functions, not a comparison of the 50 terms the entry publishes – those enter only in fixing the 5 coefficients of P . $\square$